AP CALCULUS AB

Unit 1 • Self Check 1.3 and 1.4 • Lessons 1-5 to 1-9 • Answer key

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Answer key with worked solutions

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

Which function must have an absolute minimum on the interval?

(A) $\frac{1}{x}$ on $[1,3]$

(B) $\frac{1}{x}$ on $(0,3]$

(C) x on $(0,1)$

(D) $\ln x$ on $(0,1]$

Solution

Answer: (A).

Question 2

NO CALCULATOR

What is the average rate of change of $g(x) = x^2 - 2x$ on $[1,4]$?

(A) 2

(B) 3

(C) 9

(D) -3

Solution

Answer: (B). $\frac{8-(-1)}{4-1} = 3$.

Question 3

NO CALCULATOR

h is continuous on $[0,5]$ and its graph has a sharp corner at $x = 2$. Which theorems are guaranteed to apply on $[0,5]$?

(A) MVT only

(B) IVT, EVT and MVT

(C) IVT and EVT only

(D) None of them

Solution

Answer: (C). The MVT also needs differentiability, which fails at the corner.

Question 4 (Free response) NO CALCULATOR

f is differentiable.

x	0	2	5	7
$f(x)$	3	-1	8	8

Justify each answer with a theorem. (a) Must $f(c) = 0$ for some c in $(0,2)$? (b) Must $f'(c) = 3$ for some c in $(2,5)$? (c) Must $f'(c) = 0$ for some c in $(5,7)$? (d) Must f have an absolute maximum on $[0,7]$?

Solution

(a) Yes, IVT: continuous (because differentiable) and 0 is between 3 and -1. (b) Yes, MVT: $\frac{8-(-1)}{3} = 3$. (c) Yes, MVT: $\frac{8-8}{2} = 0$. (d) Yes, EVT: continuous on the closed interval $[0,7]$.